Continuity and logarithmic regularity of the SLE_8 trace

Abstract

We prove existence and continuity of the chordal SLE_8 trace using the Loewner equation and one-dimensional Brownian motion, without imaginary geometry or a discrete approximation. For each $\alpha \in (0, 1/4)$, the trace has modulus of continuity at most a random constant times $(\log h^{-1})^{-\alpha}$ on each compact capacity-time interval. The main estimate bounds, uniformly in time, the number of dyadic height scales at which the centered inverse map has a large normalized derivative. A two-point derivative martingale controls reverse images of small height relative to a coarser reference height. A change of measure and a Brownian occupation estimate bound the number of remaining scales. Combining this scale count with an interior derivative estimate of Alvisio and Lawler gives uniform convergence of the positive-height approximations to the trace.

1 Introduction

Let B be a standard real Brownian motion with $B_0 = 0$, and set $W_t = \sqrt{8}B_t$. We consider the chordal Loewner equation in the upper half-plane $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$:

$$\partial_t g_t(z) = \frac{2}{g_t(z) - W_t}, \qquad g_0(z) = z. \tag{1.1}$$

For $z \in \mathbb{H}$, let T_z be its swallowing time, and write $H_t = \{z \in \mathbb{H} : T_z > t\}$. The map $g_t : H_t \to \mathbb{H}$ has the hydrodynamic normalization $g_t(z) = z + 2t/z + O(|z|^{-2})$ at infinity. In particular, the time parameter is normalized by $\operatorname{hcap}(\mathbb{H} \setminus H_t) = 2t$. Define the centered inverse map

$$F_t(z) = g_t^{-1}(W_t + z), \qquad z \in \mathbb{H}. \tag{1.2}$$

These maps are defined directly from (1.1), without assuming that the Loewner evolution is generated by a curve.

Theorem 1.1 (Continuity of the SLE_8 trace)**.** *Almost surely, the limit*

$$\gamma(t) = \lim_{y \downarrow 0} F_t(iy) \tag{1.3}$$

exists locally uniformly for $t \in [0, \infty)$ and defines a continuous function $\gamma : [0, \infty) \to \overline{\mathbb{H}}$ with $\gamma(0) = 0$. For every $t \geq 0$, the domain H_t is the unbounded connected component of $\mathbb{H} \setminus \gamma([0, t])$.

The proof also gives a quantitative modulus in the capacity parameterization.

Theorem 1.2 (Logarithmic modulus of continuity)**.** *Almost surely, for every $T > 0$ and every $\alpha \in (0, 1/4)$, there is a finite random constant $C = C(T, \alpha)$ such that, for all $0 \leq s < t \leq T$,*

$$|\gamma(t) - \gamma(s)| \leq C \left[\log\left(e + \frac{1}{t - s}\right)\right]^{-\alpha}. \tag{1.4}$$

Moreover, for every such T and α,

$$\lim_{h \downarrow 0} (\log h^{-1})^{\alpha} \sup_{\substack{0 \leq s < t \leq T \\ t - s \leq h}} |\gamma(t) - \gamma(s)| = 0. \tag{1.5}$$

Thus the upper bound in (1.4) has exponent $-1/4+\varepsilon$ for every $\varepsilon \in (0,1/4)$. We do not prove the endpoint bound with exponent $-1/4$, or a matching lower bound. The logarithmic modulus here is distinct from a positive-power Hölder estimate in the capacity parameter.

Relation to earlier work. Rohde and Schramm [RS05] proved continuity of the SLE_κ trace for $\kappa \neq 8$ using derivative estimates for the inverse Loewner maps. Alvisio and Lawler [AL14] obtained critical derivative estimates at $\kappa = 8$, including the all-time estimate stated in Lemma 2.1 below. Ambrosio and Miller [AM22] gave a continuous construction of the SLE_8 curve using imaginary geometry. Kavvadias, Miller, and Schoug [KMS25, Theorem 1.3] proved the upper logarithmic modulus in Theorem 1.2, together with a matching statement for exponents greater than $1/4$, by different methods. Neither of these latter two results is an input to the present proof.

The organization by height scales was motivated by Guan's manuscript [Gua07], which was withdrawn. We use no result from that manuscript as an established input. The two-point estimate, the counting argument, and the stopping-time change of measure needed here are proved below. This document is an unpublished research note.

Outline of the proof. Put $y_k = 2^{-k}$ and $A_k(t) = y_k|F_t'(iy_k)|$. The estimate of Alvisio and Lawler gives $A_k(t) \le C_\beta k^{-\beta}$ uniformly in t, for every $\beta \in (0,1/4)$. Summing this bound over all k does not control the length of $F_t(i(0,y])$. We instead show that, in each block $N < k \le 2N$, only a small number of normalized derivatives can be larger than a rapidly decaying threshold.

To compare scales, we start a reverse Loewner flow from the finest scale with a large derivative. Either its height is much smaller than a coarser reference height, in which case a two-point martingale bounds the event, or the corresponding point of a one-dimensional diffusion lies in a prescribed interval. Under a critical change of measure this diffusion is Brownian motion. Many such visits give many successive trial intervals. On each interval, substantial occupation of a fixed neighborhood of the origin has high conditional probability, whereas a large finest-scale derivative permits only a small total occupation. This proves the scale count. The resulting uniform bound on the lengths of the vertical images proves continuity and the logarithmic modulus.

2 Inverse maps and reverse flow

We first record the analytic derivative estimate and the deterministic comparisons used to move between nearby times and heights. All deterministic Loewner terminal times in Sections 2–4 will lie in $[0,1]$. Except where explicitly described as random, the letters C, c denote finite positive deterministic constants whose values may change between occurrences. Subscripts indicate permitted parameter dependence. Throughout the paper, $y_k = 2^{-k}$.

2.1 The derivative input

Lemma 2.1 (Consequence of Alvisio–Lawler)**.** *For every $\beta \in (0,1/4)$, almost surely there is a finite random constant C_β such that, for every integer $k \ge 1$,*

$$\sup_{0\le t\le 1} y_k|F_t'(iy_k)| \le C_\beta k^{-\beta}. \tag{2.1}$$

Proof. Set $\bar g_t(z) = g_t(\sqrt{8}z)/\sqrt{8}$ and $\bar F_t(z) = \bar g_t^{-1}(B_t + z)$. Then $\partial_t \bar g_t(z) = (1/4)/(\bar g_t(z) - B_t)$, which is the normalization of [AL14]. For $0 \le j \le 4^k$, put

$$\Delta_{j,k} = 1 + y_k^{-1} \sup_{jy_k^2 \le u \le (j+1)y_k^2} |B_u - B_{jy_k^2}|, \tag{2.2}$$

and define

$$\begin{aligned} D_k &= \max_{0\le j\le 4^k} |\bar F'_{jy_k^2}(iy_k)|, \qquad \widehat D_k = \sup_{0\le t\le 1} |\bar F'_t(iy_k)|, \\ D_k^* &= \max_{0\le j\le 4^k} \Delta^4_{j,k} |\bar F'_{jy_k^2}(iy_k)|. \end{aligned} \tag{2.3}$$

Lemma 1 and the definitions following it in [AL14] give $\widehat D_k \le cD_k^*$. Their Theorem 2 gives, for each $b > 0$,

$$D_k^* \le C_b(2^k k^{-b} + (\log k)^2 D_k) \tag{2.4}$$

eventually almost surely. Fix $\beta' \in (\beta, 1/4)$. Corollary 1 of [AL14] and the Borel–Cantelli lemma imply $D_k \le 2^k k^{-\beta'}$ eventually. Apply (2.4) with $b > \beta$. This proves (2.1) in their normalization. The same deduction is made in Section 3.5 of [AL14].

The centered inverse satisfies $\bar F'_t(z) = F'_t(\sqrt{8}z)$. The height $y_k/\sqrt{8}$ differs from a dyadic height by a fixed multiplicative factor. Koebe distortion along a vertical segment transfers the estimate to (2.1). The cited proof concerns interior maps throughout and does not assume the existence of a trace. □

2.2 Deterministic comparisons

Fix a continuous driver W with $W_0 = 0$ and a terminal time $T \le 1$. For $0 \le s \le T$, define

$$V_s = W_{T-s} - W_T, \qquad \partial_s h_s(z) = -\frac{2}{h_s(z) - V_s}, \qquad h_0(z) = z. \tag{2.5}$$

Write

$$Z_s(z) = h_s(z) - V_s = X_s(z) + iY_s(z), \qquad \Phi_s(z) = |h_s'(z)|. \tag{2.6}$$

In particular,

$$F_T(z) = F_{T-s}(Z_s(z)), \qquad |F_T'(z)| = \Phi_s(z)|F'_{T-s}(Z_s(z))|. \tag{2.7}$$

Indeed, $h_s(z) = g_{T-s}(F_T(z)) - W_T$ solves (2.5). Since F_0 is the identity, taking $s = T$ also gives $\Phi_T(z) = |F_T'(z)|$.

Lemma 2.2 (Height, derivative, and real-part bounds)**.** *For the reverse flow in (2.5), with initial point $z = x + iy \in \mathbb{H}$, and each $s \in [0, T]$,*

$$y \le Y_s(z) \le \sqrt{y^2 + 4s}, \qquad \Phi_s(z) \le \frac{Y_s(z)}{y}. \tag{2.8}$$

Moreover, $\operatorname{Re} h_s(z)$ *belongs to the convex hull of* $\{x\} \cup \{V_v : 0 \le v \le s\}$*. In particular,*

$$|X_s(iy)| \le 2 \sup_{0\le v\le s} |V_v|. \tag{2.9}$$

Proof. Differentiating the imaginary part and the derivative in (2.5) gives

$$\partial_s Y_s = \frac{2Y_s}{X_s^2 + Y_s^2}, \qquad \partial_s \log \Phi_s = \frac{2(X_s^2 - Y_s^2)}{(X_s^2 + Y_s^2)^2}. \tag{2.10}$$

The first equation gives $0 \le \partial_s(Y_s^2) \le 4$, and the second is at most $\partial_s \log Y_s$. This proves (2.8). The real part of h_s moves toward V_s: its derivative is negative above V_s and positive below V_s. The stated convex-hull bound follows by the scalar comparison principle, and implies (2.9). □

We now return to the Brownian driver $W = \sqrt{8}B$. For $M \in \mathbb{N}$, let Ω_M be the event that

$$\sup_{t\le 1}|W_t| \le M, \qquad |W_t - W_s| \le M\sqrt{(t-s)\log\frac{e}{t-s}} \quad \text{for } 0 \le s < t \le 1. \tag{2.11}$$

The Brownian modulus of continuity gives $\mathbb{P}(\bigcup_{M\ge 1}\Omega_M) = 1$.

Lemma 2.3 (Comparison with a time grid)**.** *For every $M \in \mathbb{N}$, there is a deterministic constant $C_M < \infty$ such that on Ω_M, for every $k \ge 1$ and $t \in [0,1]$,*

$$A_k(t) \le C_M k^2 A_k(t_k), \qquad t_k = y_k^2\lfloor t/y_k^2\rfloor, \qquad A_k(t) = y_k|F_t'(iy_k)|. \tag{2.12}$$

Proof. Reverse from terminal time t for duration $t - t_k \le y_k^2$. By Lemma 2.2 and (2.11), the image $w = X + iY$ of iy_k satisfies

$$y_k \le Y \le \sqrt{5}\,y_k, \qquad |X| \le C_M\sqrt{k}\,y_k, \qquad \Phi_{t-t_k}(iy_k) \le \sqrt{5}. \tag{2.13}$$

For a univalent map $f : \mathbb{H} \to \mathbb{C}$, Koebe distortion, applied after the change of variable $\zeta = (z - iy)/(z + iy)$, gives

$$\frac{|f'(x+iY)|}{|f'(iy)|} \le \frac{\left((x/y)^2 + (Y/y+1)^2\right)^2}{(Y/y)^3}. \tag{2.14}$$

When $1 \le Y/y \le \sqrt{5}$, the right-hand side is at most $C(1 + |x|/y)^4$. Apply this to $f = F_{t_k}$ and use (2.7) and (2.13). □

3 Derivative martingales

We use two estimates for the reverse flow. The first turns a large derivative into a restriction on the occupation time of a Brownian motion. It uses the critical martingale and change of variable from [AL14, Section 3.2]; we include the construction at a bounded stopping height. The second bounds the probability that derivatives at two different heights are both large. Both estimates will be applied at deterministic terminal times.

3.1 A change of measure at one point

Fix $T \in [0,1]$ and $y \in (0,1]$, and start the reverse flow from iy. Since the reversed increments $V_s = W_{T-s} - W_T$ have the law of $\sqrt{8}$ times a Brownian motion, we may first work with a reverse flow driven by a Brownian motion defined for all nonnegative times. Set $u_0 = \log y$ and use the increasing height parameter $u = \log Y_s$. In this parameter, write

$$J_u = \operatorname{arsinh}(X_s/Y_s), \qquad \widehat{\Phi}_u = \Phi_s, \qquad \tau_T = \log Y_T, \qquad U = \log\sqrt{5}. \tag{3.1}$$

We write $(\mathcal{F}_u)$ for the usual augmented filtration of the height-parameter process.

Lemma 3.1 (Critical change of measure)**.** *For each $T \in [0,1]$ and $y \in (0,1]$, the height-parameter process has a construction on the deterministic interval $[u_0, U]$ for which τ_T is a bounded stopping time. There is a probability measure $\mathbb{Q}$ on this interval under which $(J_{u_0+v})_{0\le v\le U-u_0}$ is a Brownian motion of variance rate 4, started at 0. For every $a \in (0,1)$ and $E \in \mathcal{F}_{\tau_T}$ with $E \subset \{y\Phi_T \ge a\}$,*

$$\mathbb{P}(E) \le y^2 a^{-2}\mathbb{Q}(E). \tag{3.2}$$

On $\{y\Phi_T \geq a\}$,

$$\tau_T \geq \log a, \qquad \int_{u_0}^{\tau_T} \mathbf{1}_{\{|J_u|\leq 1\}}\, du \leq C_{\mathrm{occ}}(1+\log(1/a)), \tag{3.3}$$

where C_{occ} *is a universal deterministic constant.*

Proof. Itô's formula, with $du = 2(X_s^2+Y_s^2)^{-1}ds$, gives

$$dJ_u = -4\tanh J_u\, du + 2\, d\beta_u, \qquad d\log\widehat{\Phi}_u = (1-2\operatorname{sech}^2 J_u)\, du. \tag{3.4}$$

Here β is a Brownian motion indexed from u_0, and $J_{u_0}=0$, $\widehat{\Phi}_{u_0}=1$. In particular,

$$y\widehat{\Phi}_u = e^u \exp\left(-2\int_{u_0}^{u} \operatorname{sech}^2 J_v\, dv\right). \tag{3.5}$$

To justify the height parameter up to the deterministic value U, construct the first SDE in (3.4) on $[u_0, U]$. Its coefficients are globally Lipschitz. Define

$$\mathfrak{s}(u) = \frac{1}{2}\int_{u_0}^{u} e^{2v}\cosh^2 J_v\, dv, \qquad Y = e^u, \qquad X = e^u \sinh J_u. \tag{3.6}$$

This is a finite, continuous, strictly increasing time change, and

$$\mathfrak{s}(U) \geq \frac{5-y^2}{4} \geq 1. \tag{3.7}$$

Itô's formula gives $dX = -X\, du + 2e^u \cosh J_u\, d\beta_u$. After inversion of (3.6), the drift is $-2X(X^2+Y^2)^{-1}ds$ and the martingale part has quadratic variation $8\,ds$. Also $dY = 2Y(X^2+Y^2)^{-1}ds$. Lévy's characterization and uniqueness of this SDE identify the process with the reverse Loewner flow, with a harmless change of Brownian sign. The second equation in (3.4) agrees with the derivative equation in (2.10). Thus

$$\tau_T = \inf\{u \in [u_0, U] : \mathfrak{s}(u) \geq T\} \tag{3.8}$$

is a stopping time bounded by U, and $\widehat{\Phi}_{\tau_T} = \Phi_T$.

Set $\mathcal{M}_u = \widehat{\Phi}_u^2 \cosh J_u$. A further application of Itô's formula gives

$$\frac{d\mathcal{M}_u}{\mathcal{M}_u} = 2\tanh J_u\, d\beta_u, \qquad \mathcal{M}_{u_0} = 1. \tag{3.9}$$

The integrand has absolute value at most 2. Novikov's criterion on the deterministic interval $[u_0, U]$ therefore implies that $\mathcal{M}$ is a true martingale. Define $\mathbb{Q}$ by $d\mathbb{Q} = \mathcal{M}_U\, d\mathbb{P}$. Girsanov's theorem cancels the drift in (3.4), so under $\mathbb{Q}$ the process J is Brownian motion of variance rate 4.

For $E \in \mathcal{F}_{\tau_T}$, optional sampling gives $\mathbb{Q}(E) = \mathbb{E}[\mathcal{M}_{\tau_T}\mathbf{1}_E]$. On $\{y\Phi_T \geq a\}$ we have $\mathcal{M}_{\tau_T} \geq \Phi_T^2 \geq a^2/y^2$, proving (3.2). Finally, (3.5) at τ_T gives $\tau_T \geq \log a$ and

$$2\int_{u_0}^{\tau_T} \operatorname{sech}^2 J_v\, dv \leq U + \log(1/a). \tag{3.10}$$

Since $\operatorname{sech}^2 J_v \geq \operatorname{sech}^2(1)$ whenever $|J_v| \leq 1$, this implies (3.3). □

3.2 Two-point derivatives

The estimate below is useful when the reverse image of a fine point has height much smaller than a coarser reference height. The gain is a positive power of the ratio of these two heights.

Lemma 3.2 (Two-point derivative estimate)**.** *For every $M \in \mathbb{N}$ and $c_0 > 0$, there is a deterministic constant $C_{M,c_0} < \infty$ such that, for every deterministic $t \in [0,1]$, $a \in (0,1)$, $y \in (0,1]$, $v \in (0, y/2]$, and $m \in \mathbb{Z}$ with $|m|v \le 1$,*

$$\mathbb{P}\left[\Omega_M \cap \{y|F_t'(iy)| \ge a,\ v|F_t'((m+i)v)| \ge c_0 a\}\right] \le C_{M,c_0} a^{-26/9} y^2 (v/y)^{16/9} (1+m^2)^{1/3}. \quad (3.11)$$

Proof. Consider two trajectories of the same reverse Brownian flow, with initial points $z_1, z_2 \in \mathbb{H}$ and $z_1 \ne z_2$. Write $Z_j = X_j + iY_j$ and $\Phi_j = |h_s'(z_j)|$, suppressing s in these quantities. For $b \in \mathbb{R}$, put

$$\rho(b) = 6b - 4b^2, \qquad \lambda(b) = 8b - 8b^2. \quad (3.12)$$

The reverse SDE and Itô's formula show that

$$\Phi_j^{\rho(b)} Y_j^{\lambda(b)-\rho(b)} (1 + X_j^2/Y_j^2)^b \quad (3.13)$$

is a local martingale with relative Brownian coefficient $-2\sqrt{8}\, bX_j/|Z_j|^2$. For completeness, writing this expression as $\Phi_j^{\rho} Y_j^{\lambda-\rho-2b}(X_j^2+Y_j^2)^b$, its relative drift, multiplied by $(X_j^2+Y_j^2)^2$, has X_j^2 and Y_j^2 coefficients

$$2\lambda - 16b + 16b^2 \quad \text{and} \quad 2\lambda - 4\rho + 8b, \quad (3.14)$$

respectively. Both vanish for (3.12).

Choose $b_1 = 1/6$ and $b_2 = 1/3$. The corresponding derivative exponents are $8/9$ and $14/9$, and both powers of Y_j in (3.13) are $2/9$. The common real Brownian terms cancel in $Z_1 - Z_2$ and $Z_1 - \overline{Z}_2$, so

$$d\log(|Z_1 - Z_2|\,|Z_1 - \overline{Z}_2|) = \frac{4X_1X_2}{|Z_1|^2|Z_2|^2}\, ds. \quad (3.15)$$

The product of the two single-point factors has relative cross drift $32 b_1 b_2 X_1 X_2/(|Z_1|^2|Z_2|^2)$. It follows that

$$\mathcal{N}_s = \frac{\Phi_1^{8/9}\Phi_2^{14/9}(Y_1Y_2)^{2/9}(1+X_1^2/Y_1^2)^{1/6}(1+X_2^2/Y_2^2)^{1/3}}{\left(|Z_1 - Z_2|\,|Z_1 - \overline{Z}_2|\right)^{4/9}} \quad (3.16)$$

is a positive local martingale, since $8b_1b_2 = 4/9$. The denominators never vanish: h_s is injective and both Z_j have positive imaginary part. In particular, $\mathbb{E}\mathcal{N}_t \le \mathcal{N}_0$.

Take $z_1 = iy$ and $z_2 = (m+i)v$. The initial distances satisfy $|z_1 - z_2| \ge y/2$ and $|z_1 - \overline{z}_2| \ge y$. Hence

$$\mathcal{N}_0 \le C y^{-4/9} (v/y)^{2/9} (1+m^2)^{1/3}. \quad (3.17)$$

Use the reversed original driver at terminal time t to realize these trajectories. On Ω_M, Lemma 2.2, the bound $|m|v \le 1$, and $\sup_{s\le t}|V_s| \le 2M$ bound both output distances in (3.16) by a constant depending only on M. On the event in (3.11), the derivative identity at $s = t$ and (2.8) give $Y_1 \ge a$ and $Y_2 \ge c_0 a$. Therefore,

$$\mathcal{N}_t \ge c_{M,c_0} a^{26/9} y^{-8/9} v^{-14/9}. \quad (3.18)$$

Combine (3.17)–(3.18) with $\mathbb{E}\mathcal{N}_t \le \mathcal{N}_0$. The use of Ω_M only supplies a pointwise lower bound for this nonnegative random variable; it does not require Ω_M to belong to the reverse filtration. □

3.3 Brownian occupation

We will apply the following estimate repeatedly after the change of measure in Lemma 3.1.

Lemma 3.3 (Occupation during a long interval)**.** *There is a universal constant $C_{\mathrm{tr}} < \infty$ such that, for every $r \geq 2$, a Brownian motion J of variance rate 4 satisfies*

$$\sup_{|x|\leq 2r} \mathbb{P}_x \left[\int_0^{2r^5} \mathbf{1}_{\{|J_u|\leq 1\}}\, du \leq r \right] \leq \frac{C_{\mathrm{tr}}}{r}. \tag{3.19}$$

Proof. Let η be the first exit from $(-r^2, r^2)$. The probability of exit within a time interval of length r^4 is bounded below by a universal positive constant, uniformly over starting points in this interval. Brownian scaling and the Markov property therefore give

$$\sup_{|x|\leq 2r} \mathbb{P}_x[\eta > 2r^5] \leq Ce^{-cr}. \tag{3.20}$$

Set $I = \int_0^\eta \mathbf{1}_{\{|J_u|\leq 1\}}\, du$. Markov's inequality gives $\mathbb{P}_x[I \leq r] \leq e\, \mathbb{E}_x[e^{-I/r}]$. The function $f(x) = \mathbb{E}_x[e^{-I/r}]$ solves

$$2f'' = r^{-1}\mathbf{1}_{[-1,1]} f \quad \text{on } (-r^2, r^2), \qquad f(-r^2) = f(r^2) = 1. \tag{3.21}$$

Equivalently, this identification follows by applying Itô's formula to the bounded C^1, piecewise C^2 solution of (3.21), stopped at η. With $\theta = (2r)^{-1/2}$, that solution is

$$f(x) = \frac{\begin{cases} \cosh(\theta x), & |x| \leq 1, \\ \cosh\theta + (|x|-1)\theta \sinh\theta, & 1 \leq |x| \leq r^2 \end{cases}}{\cosh\theta + (r^2-1)\theta\sinh\theta}. \tag{3.22}$$

For $|x| \leq 2r$, the numerator is bounded above by a universal constant, whereas the denominator is at least cr. Hence $\mathbb{P}_x[I \leq r] \leq C/r$. If $\eta \leq 2r^5$, then I is at most the occupation accumulated by time $2r^5$. Combining this observation with (3.20) proves (3.19). □

4 Counting large derivatives

We now compare all the height scales in one block. Selecting the finest large derivative gives a single reverse trajectory against which the coarser scales can be measured. The time grids are nested, so this selection reduces to a union over deterministic terminal times.

Proposition 4.1 (Number of large scales)**.** *Fix $\delta \in (0, 1/10)$. For $n \geq 1$, set*

$$N = 2^n, \qquad r = N^\delta, \qquad a = \exp(-r/n^2), \tag{4.1}$$

and, for $t \in [0,1]$, define

$$\mathcal{I}_n(t) = \{k \in \mathbb{Z} : N < k \leq 2N,\ A_k(t_k) \geq a\}, \qquad t_k = 4^{-k}\lfloor 4^k t \rfloor. \tag{4.2}$$

Almost surely there is a finite random $n_0 = n_0(\delta)$ such that, for every $n \geq n_0$,

$$\sup_{t\in[0,1]} |\mathcal{I}_n(t)| \leq r^7. \tag{4.3}$$

The parameter r determines the permitted discrepancy between logarithmic heights. The choice $\log(1/a) = r/n^2 = o(r)$ keeps the occupation allowance small compared with r. The factor a^{-2} in (3.2) is $\exp(2r/n^2)$; the accompanying factor y^2 will cancel the terminal-time grid count. After excluding small image heights, more than r^7 large scales will give at least $\lceil r \rceil$ trial intervals of length $2r^5 = o(N)$.

Proof. Fix $M \in \mathbb{N}$. All deterministic comparisons below are made on Ω_M. Their validity is required only for n larger than a deterministic threshold depending on M and δ. We bound two exceptional events by summable sequences and then apply Borel–Cantelli.

Step 1: Selection of a terminal time. Suppose $\mathcal{I}_n(t)$ is nonempty, and put

$$k_* = \max \mathcal{I}_n(t), \qquad T = t_{k_*}, \qquad y_* = 2^{-k_*}. \tag{4.4}$$

For every $k \le k_*$, nesting of the grids gives

$$t_k = 4^{-k} \lfloor 4^k T \rfloor. \tag{4.5}$$

Indeed, $t \in [T, T + 4^{-k_*})$, and this interval cannot cross a grid point of spacing 4^{-k} in its interior. Thus (k_*, T) determines all the relevant coarser times, independently of the choice of t in that interval.

Run the reverse flow from (T, iy_*). For $k \in \mathcal{I}_n(t)$, let $s_k = T - t_k$ and $w_k = Z_{s_k}(iy_*)$. By Lemma 2.2 and the localization event,

$$0 \le s_k \le y_k^2, \qquad y_* \le Y_{s_k} \le \sqrt{5}\, y_k, \qquad |X_{s_k}| \le C_M \sqrt{N}\, y_k. \tag{4.6}$$

The last inequality follows because the reverse driver increments over this interval are at most $M y_k \sqrt{\log(e/y_k^2)}$.

Step 2: Small image heights. Let $\mathcal{U}_n$ be the event that some t and some $k \in \mathcal{I}_n(t)$, with the selection in (4.4), satisfy

$$Y_{s_k} < e^{-r} y_k. \tag{4.7}$$

Choose $v = 2^{-\ell}$ and $m \in \mathbb{Z}$ so that

$$v \le Y_{s_k} < 2v, \qquad |X_{s_k} - mv| \le v. \tag{4.8}$$

Set $d = \ell - k$. Since $Y_{s_k} \ge y_*$, we have $\ell \le k_* \le 2N$. Equations (4.6)–(4.8) imply

$$d \ge r/\log 2 - C, \qquad |m| \le C_M \sqrt{N}\, 2^d, \qquad |m| v \le 1 \tag{4.9}$$

for all sufficiently large n. Also $v \le y_k/2$ for such n.

The derivative at the finest scale is large. By (2.7) and (2.8),

$$a \le y_* |F_T'(iy_*)| = y_* \Phi_{s_k}(iy_*) |F_{t_k}'(w_k)| \le Y_{s_k} |F_{t_k}'(w_k)|. \tag{4.10}$$

The points w_k and $(m+i)v$ have bounded hyperbolic distance by (4.8). Koebe distortion therefore gives, for a universal $c_0 > 0$,

$$v |F_{t_k}'((m+i)v)| \ge c_0 a. \tag{4.11}$$

The fact that $k \in \mathcal{I}_n(t)$ supplies the other derivative inequality in Lemma 3.2.

We sum (3.11) over the possible coarse indices k, their grid times, and the values of ℓ, m in (4.9). For each k, there are at most $2y_k^{-2}$ grid times in $[0,1]$. Furthermore,

$$\sum_{|m|\le C_M\sqrt{N}\,2^d} (1+m^2)^{1/3} \le C_M N^{5/6} 2^{5d/3}. \tag{4.12}$$

The time count cancels the factor y_k^2 in (3.11), and $2^{-16d/9}2^{5d/3} = 2^{-d/9}$. Consequently,

$$\begin{aligned}\mathbb{P}(\Omega_M \cap \mathcal{U}_n) &\le C_M N^{11/6} a^{-26/9} \sum_{d \ge r/\log 2 - C} 2^{-d/9} \\ &\le C_M N^{11/6} \exp\left(\frac{26r}{9n^2} - \frac{r}{9}\right) \le C_M N^2 e^{-cr}.\end{aligned} \tag{4.13}$$

This is summable over n. No additional union over (k_*, T) is taken here: the existence of a small image height already implies one of the coarse-time two-point events in this sum.

Step 3: Many remaining scales force many visits. Suppose $|\mathcal{I}_n(t)| > r^7$ and $\mathcal{U}_n$ does not occur. For the reverse flow selected in (4.4), use the height parameter from (3.1), and set $u_k = \log Y_{s_k}$. Since (4.7) fails,

$$-k\log 2 - r \le u_k \le -k\log 2 + \log\sqrt{5}. \tag{4.14}$$

Together with (4.6), this gives

$$|J_{u_k}| \le r + \log(C_M\sqrt{N}) + C \le 2r. \tag{4.15}$$

The final inequality holds eventually since $r = N^\delta$ dominates $\log N$. Also

$$u_0 = \log y_* \le u_k \le u_{\mathrm{end}}, \qquad u_{\mathrm{end}} = -N\log 2 + \log\sqrt{5}. \tag{4.16}$$

Set $L = r^5$, and define successive entrance times

$$\sigma_1 = u_0, \qquad \sigma_{j+1} = \inf\{u \ge \sigma_j + 2L : |J_u| \le 2r\}. \tag{4.17}$$

The diffusion can be continued beyond U to define these times. All events to which we apply the change of measure below require the relevant intervals to finish before $\tau_T \le U$.

Every visit to $[-2r, 2r]$ during $[\sigma_j, \sigma_{j+1})$ lies in $[\sigma_j, \sigma_j + 2L]$. By (4.14), an interval of height-parameter length $2L$ can contain u_k for at most

$$\frac{2L + r + \log\sqrt{5}}{\log 2} + 2 \le C(L + r + 1) \tag{4.18}$$

distinct indices k. Set $m_r = \lceil r \rceil$. If $\sigma_{m_r} > u_{\mathrm{end}}$, all the times in (4.15) lie in the first $m_r - 1$ such intervals. The number of corresponding indices would then be at most $Cr(L + r + 1) = O(r^6)$, contradicting $|\mathcal{I}_n(t)| > r^7$ for large n. Hence

$$\sigma_{m_r} \le u_{\mathrm{end}}. \tag{4.19}$$

Since the finest derivative is large, Lemma 3.1 gives

$$\tau_T \ge \log a = -r/n^2, \qquad \int_{u_0}^{\tau_T} \mathbf{1}_{\{|J_u| \le 1\}}\, du \le C_{\mathrm{occ}}(1 + r/n^2) \le r \tag{4.20}$$

for large n. Our restriction on δ implies $5\delta < 1$, so

$$u_{\mathrm{end}} + 2L < \log a. \tag{4.21}$$

Thus the first m_r intervals $[\sigma_j, \sigma_j + 2L]$ all finish before τ_T, and each has occupation of $[-1, 1]$ at most r.

Step 4: Probability of many visits. Fix a deterministic pair (k_*, T) with $N < k_* \le 2N$ and T on the grid of spacing y_*^2. Let $E_{k_*,T}$ be the event that

$$\begin{aligned} y_* \Phi_T(iy_*) \ge a, \qquad \sigma_{m_r} \le u_{\mathrm{end}}, \qquad \sigma_{m_r} + 2L \le \tau_T, \\ \int_{\sigma_j}^{\sigma_j + 2L} \mathbf{1}_{\{|J_u| \le 1\}}\, du \le r \quad \text{for } 1 \le j \le m_r. \end{aligned} \tag{4.22}$$

The completion condition makes this an $\mathcal{F}_{\tau_T}$-measurable event. It also ensures that all displayed integrals have finite limits of integration.

Under $\mathbb{Q}$, continue J after U using independent Brownian increments. This does not change $E_{k_*,T}$. Brownian recurrence implies that every σ_j is finite, $\mathbb{Q}$-almost surely. Drop all conditions in $E_{k_*,T}$ except the occupation bounds in the first m_r trial intervals. At σ_j, the process lies in $[-2r, 2r]$; this includes $\sigma_1 = u_0$, where it is 0. The occupation in the jth interval is measurable by $\sigma_j + 2L \le \sigma_{j+1}$. Lemma 3.3 and successive applications of the strong Markov property give

$$\mathbb{Q}(E_{k_*,T}) \le (C_{\mathrm{tr}}/r)^{m_r} \le e^{-cr \log r}. \tag{4.23}$$

By (3.2),

$$\mathbb{P}(E_{k_*,T}) \le y_*^2 a^{-2} e^{-cr \log r}. \tag{4.24}$$

Steps 1 and 3 show that, on $\Omega_M \setminus \mathcal{U}_n$, the existence of t with $|\mathcal{I}_n(t)| > r^7$ implies $E_{k_*,T}$ for some deterministic pair in the preceding paragraph. The localization event is dropped in this inclusion before applying (3.2); no assertion about its probability under $\mathbb{Q}$ is used. For each k_*, there are at most $2y_*^{-2}$ possible terminal times. Summing (4.24) over the pairs gives

$$\begin{aligned} \mathbb{P}\left[\Omega_M \cap \mathcal{U}_n^c \cap \{\exists t \in [0,1] : |\mathcal{I}_n(t)| > r^7\}\right] \\ \le 2N a^{-2} e^{-cr \log r}. \end{aligned} \tag{4.25}$$

This is summable over n, because $a^{-2} = \exp(2r/n^2)$ and $r = 2^{n\delta}$.

The events involving an arbitrary t above are measurable finite unions: the finite collection of grid values $(t_k)_{N<k\le 2N}$ is constant on each grid interval of spacing 4^{-2N}, with the endpoint 1 treated separately. Apply Borel–Cantelli to (4.13) and (4.25). On Ω_M, (4.3) holds for all sufficiently large n. Taking the countable union over M proves the proposition. □

5 Radial convergence and the trace

The scale count gives a summable bound when combined with Lemma 2.1. We retain the rate of the sum, since it will also control the modulus of continuity. In the pathwise estimates of this section, an unsubscripted C may depend on the Brownian sample and the fixed exponents.

Proposition 5.1 (Uniform radial variation)**.** *For every $\alpha \in (0, 1/4)$, almost surely there is a finite random constant C_α such that, for every $y \in (0, 1/2]$,*

$$\sup_{0 \le t \le 1} \int_0^y |F_t'(iv)|\, dv \le C_\alpha [\log(e/y)]^{-\alpha}. \tag{5.1}$$

The integrals are improper integrals of nonnegative functions.

Proof. Choose $\beta \in (\alpha, 1/4)$ and

$$0 < \delta < \min\{1/10, (\beta - \alpha)/7\}, \qquad q = \beta - 7\delta > \alpha. \tag{5.2}$$

Work on the probability-one event on which Lemma 2.1, Proposition 4.1, and some Ω_M hold. Retain the notation $N = 2^n$, $r = N^\delta$, and $a = \exp(-r/n^2)$ from Proposition 4.1.

For every sufficiently large n and every $t \in [0,1]$, at most r^7 indices in $(N, 2N]$ belong to $\mathcal{I}_n(t)$. For those indices use (2.1) at the actual time t. For the other indices use (2.12) and $A_k(t_k) < a$. This gives

$$\begin{aligned}\sup_{t\le 1} \sum_{N<k\le 2N} A_k(t) &\le C_M N^3 a + C_\beta r^7 N^{-\beta} \\ &= C_M N^3 e^{-N^\delta/n^2} + C_\beta N^{-q} \le C N^{-q}.\end{aligned} \tag{5.3}$$

Here the last inequality holds eventually since N^δ/n^2 dominates every fixed multiple of $\log N$.

For every sufficiently large integer K, choose n so that $2^n < K \le 2^{n+1}$. Summing (5.3) over the blocks with lower endpoints 2^j, $j \ge n$, gives

$$\sup_{t\le 1} \sum_{k\ge K} A_k(t) \le C \sum_{j\ge n} 2^{-jq} \le C K^{-q}. \tag{5.4}$$

For a dyadic vertical interval, fixed-factor Koebe distortion gives

$$\int_{y_{k+1}}^{y_k} |F_t'(iv)|\, dv \le C y_k |F_t'(iy_k)| = C A_k(t). \tag{5.5}$$

Summing (5.5) and using (5.4) bounds the integral up to y_K by CK^{-q}, uniformly in t. Monotonicity of the integral extends the estimate to all intermediate heights. The finitely many larger height intervals are absorbed into the random constant: (2.8) and $\Phi_t(iy_k) = |F_t'(iy_k)|$ give $\sup_{t\le 1} A_k(t) \le \sqrt{y_k^2 + 4} \le \sqrt{5}$ for every $k \ge 1$. As $q > \alpha$ and K is comparable to $\log(e/y_K)$, this proves (5.1). □

Proof of Theorem 1.1. For $0 < y' < y \le 1/2$,

$$\sup_{t\le 1} |F_t(iy) - F_t(iy')| \le \sup_{t\le 1} \int_0^y |F_t'(iv)|\, dv. \tag{5.6}$$

Proposition 5.1 shows that the right-hand side tends to zero as $y \downarrow 0$. Hence $F_t(iy)$ has a limit $\gamma(t)$ uniformly over $t \in [0,1]$. For each fixed $y > 0$, the function $t \mapsto F_t(iy)$ is continuous: one may use the reverse equation (2.5), whose solutions stay at imaginary part at least y, and continuous dependence on the terminal time and the continuous driver. Therefore γ is continuous, and $\gamma(0) = 0$.

Brownian scaling applies the same conclusion on each deterministic interval $[0, j]$, $j \in \mathbb{N}$. Intersecting the resulting probability-one events gives a continuous limit on $[0, \infty)$, with locally uniform convergence. The deterministic trace criterion [RS05, Theorem 4.1] states that for a continuous driver, existence and continuity of all vertical limits in (1.3) imply that H_t is the unbounded connected component of $\mathbb{H} \setminus \gamma([0,t])$. Its hypotheses have just been verified. □

Proof of Theorem 1.2. Fix $\alpha \in (0, 1/4)$. First work on $[0,1]$, on the probability-one event where Theorem 1.1, Proposition 5.1 for this α, and some Ω_M hold. Taking the limit at the lower endpoint of the integral in (5.1) gives

$$\sup_{t\le 1} |\gamma(t) - F_t(iy)| \le C_\alpha [\log(e/y)]^{-\alpha}, \qquad 0 < y \le 1/2. \tag{5.7}$$

Let $0 \le s < t \le 1$, set $h = t - s$, and take $y = h^{1/4}$. Reverse from terminal time t for physical duration h, starting at iy, and let $w = X + iY$ be the resulting centered point. The composition identity gives $F_t(iy) = F_s(w)$. By Lemma 2.2 and the Brownian modulus,

$$|X| \le 2M\sqrt{h\log(e/h)}, \qquad 0 \le Y - y \le 2h/y. \tag{5.8}$$

Consequently,

$$\frac{|w - iy|}{y} \le 2Mh^{1/4}\sqrt{\log(e/h)} + 2h^{1/2}. \tag{5.9}$$

For all sufficiently small h, depending only on M, the point w lies in $B(iy, y/2)$. Koebe distortion on this disk and integration along the straight segment give

$$|F_t(iy) - F_s(iy)| \le C|F_s'(iy)|\,|w - iy|. \tag{5.10}$$

Applying (2.8) to the reverse flow from terminal time s gives the deterministic bound

$$y|F_s'(iy)| \le \sqrt{y^2 + 4s} \le \sqrt{5}. \tag{5.11}$$

It follows that

$$|F_t(iy) - F_s(iy)| \le C_M\left(h^{1/4}\sqrt{\log(e/h)} + h^{1/2}\right). \tag{5.12}$$

The right-hand side is smaller, for small h, than a constant times $(\log(e/h))^{-\alpha}$. Combining (5.7) at times s, t with (5.12), and using $\log(1/y) = \frac{1}{4}\log(1/h)$, proves (1.4) for sufficiently small h on $[0,1]$. The remaining values of h are absorbed into the constant, since $\gamma([0,1])$ is bounded.

Apply Brownian scaling on each interval $[0, j]$, $j \in \mathbb{N}$, and take a countable intersection over these intervals and rational exponents in $(0, 1/4)$. For an arbitrary $\alpha \in (0, 1/4)$ choose a larger rational exponent still below $1/4$. Its bound implies (1.4) for α, and also (1.5), because

$$(\log h^{-1})^{\alpha}\left[\log\left(e + h^{-1}\right)\right]^{-\alpha'} \longrightarrow 0 \quad \text{whenever } \alpha < \alpha' < 1/4. \tag{5.13}$$

Every finite T is contained in an integer interval, which proves the simultaneous assertions. □

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